

Dr. Weymar Andrés Astaiza Sulez

Professor of Mathematics
Universidad del Cauca, Colombia

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1 Short Bio

Weymar Andrés Astaiza Sulez is a mathematician and researcher currently serving as a Professor at the Universidad del Cauca, Colombia. He received his Ph.D. in Mathematics from the University of São Paulo, Brazil, in 2024, and his Master's degree in Mathematics from University of Los Andes, Colombia. His research interests include partial differential equations, semigroup theory, pseudodifferential operators, functional analysis, and computational mathematics. He has published research articles in mathematical analysis and is currently working on pseudodifferential operators and related problems. He is also involved in the organization of mathematical activities through UMALCA–ICMAM and serves as an associate researcher in the ICMAM Latin America Research Group in Analysis and PDE.

Research Interests

Keywords: Partial differential equations; pseudodifferential operators; functional analysis; analytic semigroups; operator theory; applied mathematics; computational mathematics.

Academic Positions

Professor March 2023–Present
Universidad del Cauca, Colombia

- Teaching Computational Algorithms and undergraduate Mathematics courses.
- Research in Partial Differential Equations, Pseudodifferential Operators, Functional Analysis, and related areas of Analysis.

Professor August 2017–May 2018
Universidad de los Andes, Colombia

Professor April–December 2016
SENA Regional Cauca, Colombia

Professor February–April 2016
SENA Regional Valle, Colombia

2 Education

Ph.D. in Applied Mathematics February 2024
University of São Paulo, Brazil

Dissertation: *Parabolic Equations on Conic Manifolds*

Advisor: Prof. Pedro Tavares Paes Lopes

M.Sc. in Mathematics July 2017
Universidad de Los Andes, Bogotá, Colombia

Degree Project: *The Monte Carlo Method and Its Applications to Elliptic Equations*

Advisor: Prof. Jean Carlos Cortissoz Iriarte

B.Sc. in Mathematics September 2013
Universidad del Cauca, Colombia

Degree Project: *The Babuška–Brezzi Theory and Its Applications to the Stokes and Darcy Equations*

Advisor: Prof. Ramiro Acevedo

Scholarships and Funding

CAPES Doctoral Scholarship

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil.

Doctoral studies in Applied Mathematics at the University of São Paulo.

Graduate Scholarship

Universidad de los Andes, Colombia.

Scholarship supporting Master's studies in Mathematics.

Research Experience

Doctoral Research – Partial Differential Equations

University of São Paulo, Brazil

2019–2024

Research on parabolic partial differential equations on conic manifolds, with emphasis on functional analytic methods and operator theory.

Master's Research – Numerical Methods for PDEs

Universidad de los Andes, Colombia

2015–2017

Research on Monte Carlo methods and their applications to elliptic partial differential equations.

Undergraduate Research – Numerical Analysis and PDEs

Universidad del Cauca, Colombia

2012–2013

Research on Babuška–Brezzi theory and its applications to the Stokes and Darcy problems.

Collaborative Research – Graph Theory

Research on symmetric tensor powers of graphs, resulting in a publication in *Scientia Series A: Mathematical Sciences*.

Publications

1. Astaiza Sulez, W. A., Barrios, A. J., Chimal-Dzul, H., Garcia, S. R., Lopez de la Luz, J., Moll, V. H., Puig, Y., & Villamizar, D.
Symmetric Tensor Powers of Graphs.
Scientia Series A: Mathematical Sciences, Vol. 36 (2026), pp. 13–35.
2. Astaiza Sulez, W. A. & Lopes, P. T. P.
Sectorial Pseudodifferential Operators in $L_1(\mathbb{R}^n)$ and $BUC(\mathbb{R}^n)$.
In *Analysis and PDE in Latin America (ICMAM 2024)*, Trends in Mathematics, Vol. 15, pp. 25–31.
Springer, 2026.
First published online: January 2, 2026.
3. Book: Quintero, J. R., Delgado, J., Gallego, F. A., Astaiza Sulez, W. A., Cardona Sánchez, D., eds.
Analysis and PDE in Latin America: ICMAM 2026.
Trends in Mathematics, Springer Birkhäuser.
Forthcoming, 2027.

Manuscripts in Preparation

1. Astaiza Sulez, W. A. & Lopes, P. T. P.
Manuscript on Partial Differential Equations.
Manuscript in preparation; planned for submission in 2026.

Conference Presentations

1. Astaiza Sulez, W. A.

Sectorial Pseudodifferential Operators on $L_1(\mathbb{R}^n)$ and $BUC(\mathbb{R}^n)$.

XXIV Congreso Colombiano de Matemáticas, Sociedad Colombiana de Matemáticas, Universidad Tecnológica de Pereira, Pereira, Colombia, June 2–6, 2025.

Oral Presentation.

Teaching Experience

Professor – Computational Algorithms and Mathematics

Universidad del Cauca, Colombia

March 2023–Present

- Teaching Computational Algorithms.
- Teaching undergraduate Mathematics courses.
- Development of lectures, computational exercises, programming assignments, and mathematical applications.
- Use of Python and computational tools for mathematical problem solving and algorithmic development.

Professor

Universidad de los Andes, Colombia

August 2017–May 2018

- Teaching undergraduate Mathematics courses.

Graduate Teaching Assistant

Universidad de los Andes, Colombia

January 2014–May 2017

- Assisted in undergraduate courses including Linear Algebra, Vector Calculus, and Integral Calculus.
- Delivered a lecture in Linear Algebra.
- Assisted instructors with course preparation, problem sessions, and academic activities.

Professor

SENA Regional Cauca, Colombia

April–December 2016

- Teaching Mathematics and related quantitative subjects.

Professor

SENA Regional Valle, Colombia

February–April 2016

- Teaching Mathematics and related quantitative subjects.

Graduate and Research Assistantships

Ph.D. Graduate Assistant

University of São Paulo, Brazil

June 2021–November 2022

Graduate Assistant

Universidad de los Andes, Colombia

January–May 2017

Graduate Assistant

Universidad de los Andes, Colombia

January 2014–May 2015

Research Assistant

Universidad de los Andes, Colombia

June–July 2015

3 Academic Service

Organizing Committee Member UMALCA–ICMAM Latin American Mathematics Seminar 2025–Present

- Member of the organizing committee, contributing to the organization of scientific activities and academic events.

Associate Researcher ICMAM Latin America – Research Group in Analysis and PDE 2025–Present

- Research sponsor: Prof. Duván Cardona Sánchez, King Fahd University of Petroleum and Minerals, College of Computing and Mathematics, Dhahran, Saudi Arabia.

Editorial Activities

Editor

Analysis and PDE in Latin America: ICMAM 2026

Springer / Birkhäuser

2026–Present

Languages

Spanish: Native

English: Fluent

Portuguese: Advanced

References

Prof. Pedro Tavares Paes Lopes

Associate Professor

University of São Paulo, São Paulo, Brazil

Email: pplopes@icmc.usp.br | pplopes@ime.usp.br

Ph.D. Advisor

Prof. Jean Carlos Cortissoz Iriarte

Associate Professor

Universidad de los Andes, Bogotá, Colombia

Email: jcortiss@uniandes.edu.co

M.Sc. Advisor

Prof. René Joaquín Meziat Vélez

Associate Professor

Universidad de los Andes, Bogotá, Colombia

Email: rmeziat@uniandes.edu.co

Former Teaching Assistant Supervisor

Prof. Duván Cardona Sánchez

Assistant Professor of Mathematics

King Fahd University of Petroleum and Minerals

College of Computing and Mathematics

Dhahran, Kingdom of Saudi Arabia

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